



UNIVERSITY OF ZIMBABWE

CONCEPT NOTE

Centralised Room Inventory and Usage Monitoring System

1. Proposed Project Title

Development of a Centralised Room Inventory and Usage Monitoring System for the University of Zimbabwe

2. Background

The University of Zimbabwe has many teaching, learning, administrative, examination and event spaces distributed across faculties, departments, schools, institutes and administrative units. These rooms include lecture rooms, laboratories, seminar rooms, boardrooms, computer laboratories, examination venues, halls and other special-purpose spaces. Each of these rooms has different characteristics such as seating capacity, furniture type, building location, available equipment, accessibility, ownership or department responsible, and suitability for lectures, examinations, meetings, workshops or public events.

At present, room information and usage schedules may be managed separately by different offices, departments or secretaries. This can result in fragmented records, double-booking, underutilisation of available spaces, difficulty in identifying free rooms, and delays in approving room bookings. The absence of a centralised digital system makes it difficult for university management, planning offices and departments to have a real-time view of room availability, capacity, utilisation and booking demand.

To address this, there is need for a **centralised room inventory and usage monitoring system** that stores all university rooms in one platform and allows authorised staff to update room details, monitor room status, and manage bookings. The system will support improved space planning, efficient scheduling, transparency in room usage, and evidence-based decision-making.

3. Problem Statement

The University of Zimbabwe currently requires an efficient way to maintain a complete and up-to-date inventory of rooms and monitor their dynamic usage for lectures, examinations, meetings, events and other institutional activities. Without a centralised digital system, room information

may be scattered across departments, making it difficult to know which rooms exist, their capacities, their current usage status, and whether they are available for booking.

This can lead to challenges such as double-booking, poor room allocation, underutilisation of some spaces, overcrowding in others, and slow communication between departments and responsible offices. There is therefore a need for a secure, centralised and user-friendly system that enables authorised staff to manage room records and update usage information, while allowing read-only users to view room availability and utilisation dashboards.

4. Aim of the Proposed System

The aim of the proposed system is to develop a **centralised digital platform for recording, managing, monitoring and reporting room inventory and room usage across the University of Zimbabwe.**

5. Objectives

The proposed system will seek to:

1. Create a centralised digital inventory of all rooms at the University of Zimbabwe, including room names, locations, capacities, types, departments and available facilities.
2. Enable authorised administrative users to insert, update and manage room records within their assigned departments or offices.
3. Support dynamic updating of room usage status, including whether a room is free, occupied, booked, under maintenance, reserved for examinations, or unavailable.
4. Provide a structured booking workflow where room booking requests can be submitted, reviewed and forwarded to responsible offices for approval.
5. Prevent double-booking by checking room availability against existing schedules and bookings.
6. Provide dashboards and reports showing room utilisation, availability, booking frequency, capacity distribution and departmental room usage.
7. Improve institutional planning by giving management and planning offices access to reliable room inventory and usage statistics.

6. Proposed System Overview

The proposed system will be a web-based platform accessible through computers and possibly mobile devices on the university network or secure online environment. It will contain a database of all rooms and their attributes. Authorised users will be able to update room information and

room usage status, while read-only viewers will be able to view approved room availability and dashboard summaries.

The system will have two main user categories:

- | | |
|---|----------------|
| 1. Admin | Users |
| These are authorised staff members who can manage data. This category may include department secretaries, faculty administrators, estates officers, examinations officers, planning officers and system administrators. Admin access can be restricted according to responsibility. For example, a department secretary may only update rooms belonging to their department, while a central administrator may manage all university rooms. | |
| 2. Read-Only | Viewers |
| These are users who can view room information, availability and dashboards but cannot edit or delete records. This category may include lecturers, heads of departments, faculty officers, planning staff, students' representatives or other approved university users. | |

Although the system will have two main user categories, the admin category can still have internal permission levels to ensure that each person only manages the rooms or functions assigned to them.

7. Key System Modules

7.1 Room Inventory Module

This module will store the official record of all rooms at the university. Each room will have a unique record and may include the following information:

- Room code or room ID
- Room name or number
- Building name
- Faculty, department or office responsible
- Campus or area
- Floor level
- Room type
- Seating capacity
- Standing capacity, where applicable
- Room furniture type
- Available equipment

- Accessibility features
- Room condition
- Ownership or responsible office
- Current operational status
- Photographs of the room, where necessary
- Remarks or special notes

Room types may include:

- Lecture room
- Seminar room
- Laboratory
- Computer laboratory
- Boardroom
- Examination venue
- Hall
- Office meeting room
- Studio
- Workshop space
- Special-purpose room

7.2 Room Usage Status Module

This module will allow authorised users to update the current or scheduled use of a room. Possible room statuses may include:

- Available
- In use
- Booked
- Reserved for lecture
- Reserved for examination
- Reserved for meeting
- Reserved for event

- Under maintenance
- Temporarily unavailable
- Closed
- Cleaning in progress

This will allow users to quickly understand whether a room can be used at a given date and time.

7.3 Booking and Scheduling Module

The booking module will allow users to request rooms for lectures, examinations, meetings, workshops, conferences, training sessions or university events. A booking request may capture:

- Name of requester
- Department or office
- Purpose of booking
- Room requested
- Expected number of participants
- Date and time
- Required equipment
- Special requirements
- Contact person
- Approval status

The system should check room availability before a booking is submitted. Where a room is already booked, the system should notify the user and suggest alternative available rooms with suitable capacity.

Possible booking statuses may include:

- Pending approval
- Approved
- Rejected
- Cancelled
- Completed
- Rescheduled

7.4 Approval Workflow Module

Some bookings may require approval from responsible offices such as department offices, faculty offices, examinations office, estates office, planning office or central administration. The system should allow booking requests to be routed to the appropriate office.

A basic workflow can be:

1. User submits booking request.
2. System checks availability and capacity.
3. Request is sent to responsible admin office.
4. Admin reviews request.
5. Booking is approved, rejected or returned for correction.
6. Requester receives notification.
7. Approved booking appears on the room schedule and dashboard.

This workflow will improve accountability and reduce informal room allocation.

7.5 Dashboard and Reporting Module

The dashboard will provide a visual summary of room usage across the university. It should allow management and authorised users to quickly understand how rooms are being used.

Possible dashboard indicators include:

- Total number of rooms in the system
- Number of available rooms
- Number of rooms currently in use
- Number of rooms booked for lectures
- Number of rooms booked for examinations
- Number of rooms under maintenance
- Room utilisation rate by department
- Room utilisation rate by building
- Most used rooms
- Least used rooms

- Rooms with high demand
- Rooms with low usage
- Capacity distribution of rooms
- Upcoming bookings
- Booking requests awaiting approval

Reports may be exported as PDF, Excel or printed summaries for planning meetings.

7.6 User Management Module

The system will include secure user accounts. Users will log in using unique credentials. The system administrator will be able to create, update, disable or delete user accounts.

The two main user categories will be:

Admin

Can add, edit and manage room data, bookings and usage information depending on assigned permissions.

User

Read-Only

Can view approved room data, schedules and dashboards but cannot change information.

Viewer

For better control, admin permissions may be scoped by department, faculty or office. For example, a secretary in the Department of Geoinformatics may only update rooms assigned to that department, while a central planning officer may view and manage all room records.

7.7 Notification Module

The system should support notifications for important actions. Notifications may be sent by email or displayed inside the system.

Examples include:

- Booking request submitted
- Booking approved
- Booking rejected
- Booking cancelled
- Room placed under maintenance
- Room status changed
- Upcoming booking reminder

- Double-booking warning
- Pending approval reminder

7.8 Audit Trail Module

To ensure accountability, the system should record all important changes made by users. The audit trail should show:

- User who made the change
- Date and time of change
- Room affected
- Type of action performed
- Previous value
- New value

This will help prevent unauthorised changes and support administrative transparency.

8. Proposed Data Fields

The system database may include the following main tables:

8.1 Room Table

- Room ID
- Room name or number
- Building
- Faculty
- Department
- Room type
- Capacity
- Floor
- Equipment available
- Condition
- Accessibility status
- Responsible office

- Current status
- Remarks

8.2 Booking Table

- Booking ID
- Room ID
- Requester name
- Department
- Purpose
- Start date and time
- End date and time
- Expected attendance
- Required equipment
- Booking status
- Approved by
- Date approved
- Remarks

8.3 User Table

- User ID
- Full name
- Email
- Department
- Role
- Permission level
- Account status

8.4 Usage Log Table

- Log ID
- Room ID
- Usage type

- Date and time
- Updated by
- Status change
- Remarks

9. Functional Requirements

The system should be able to:

1. Register all university rooms in a central database.
2. Store room capacities, types, equipment and responsible departments.
3. Allow authorised admin users to add and update room records.
4. Allow users to search for rooms by building, department, capacity, type or status.
5. Allow room usage status to be updated dynamically.
6. Allow users to submit room booking requests.
7. Check for scheduling conflicts before approving bookings.
8. Route booking requests to responsible offices.
9. Allow admins to approve, reject or cancel bookings.
10. Display room availability in calendar format.
11. Display dashboard statistics on room usage.
12. Generate reports for planning and management.
13. Maintain an audit trail of system activities.
14. Secure the system using user login and role-based access control.

10. Non-Functional Requirements

The system should be:

- **Secure:** Only authorised users should access restricted functions.
- **User-friendly:** Secretaries and administrators should be able to use the system with minimal training.
- **Reliable:** The system should be stable and available when needed.
- **Scalable:** It should support future expansion to more campuses, buildings and departments.

- **Responsive:** It should work on desktops, laptops and tablets.
- **Auditable:** Changes to important records should be traceable.
- **Maintainable:** The system should be easy to update and improve over time.
- **Centralised:** All room data should be stored in one official database.

11. Proposed Users and Access Rights

User Category	Main Users	Main Capabilities
Admin User	Department secretaries, faculty administrators, examinations office, estates office, planning office, system administrator	Add rooms, update room details, update usage status, process bookings, approve requests, generate reports, manage assigned data
Read-Only Viewer	Lecturers, management, planning viewers, approved staff, selected student representatives	View room inventory, check availability, view dashboards, view approved schedules, download allowed reports

12. Proposed Dashboard Views

The main interface should be a dashboard showing the current status of university rooms. Suggested dashboard sections include:

12.1 University-Wide Overview

This view shows total rooms, total capacity, available rooms, occupied rooms, booked rooms and unavailable rooms.

12.2 Departmental View

This view shows rooms belonging to a selected department, their capacities, current usage and booking status.

12.3 Building View

This view shows rooms grouped by building, helping users identify available spaces in specific locations.

12.4 Timetable and Calendar View

This view displays room usage by day, week or month.

12.5 Booking Requests View

This view shows pending, approved, rejected and cancelled bookings.

12.6 Maintenance and Unavailable Rooms View

This view shows rooms that are closed, under maintenance or temporarily unavailable.

13. Expected Benefits

The proposed system is expected to provide the following benefits:

- Improved visibility of all university rooms and capacities.
- Reduced double-booking of rooms.
- Faster room booking and approval processes.
- Better planning for lectures, examinations and events.
- Improved utilisation of underused spaces.
- Reliable data for infrastructure planning.
- Improved communication between departments and central offices.
- Easier monitoring of rooms under maintenance or unavailable rooms.
- Evidence-based decision-making through dashboards and reports.
- Stronger accountability through user logs and audit trails.

14. Suggested Technology Stack

The system can be developed using affordable and secure technologies. A suitable stack may include:

- **Frontend:** React, HTML, CSS, JavaScript or Bootstrap
- **Backend:** Django, Laravel, Node.js or FastAPI
- **Database:** PostgreSQL
- **Authentication:** Role-based login system
- **Reporting:** PDF and Excel export tools
- **Hosting:** University server, secure cloud server or local institutional server
- **Optional Mapping/Floor Component:** Building and room location maps can be added later if the university wants spatial navigation

For long-term institutional use, PostgreSQL is recommended because it is reliable, open-source and suitable for structured institutional records.

15. Implementation Approach

The development can be done in phases.

Phase 1: Requirements Gathering

This phase involves meeting relevant offices and departments to confirm room categories, booking rules, approval responsibilities and reporting needs.

Phase 2: System Design

This phase involves designing the database, user roles, dashboard layout, booking workflow and system interface.

Phase 3: Prototype Development

A working prototype will be created with core modules such as room inventory, user login, room status update, booking request and dashboard.

Phase 4: Data Collection and Entry

All rooms, capacities, departments, buildings and room types will be collected and entered into the system.

Phase 5: Testing and Validation

The system will be tested with sample departments to check usability, data accuracy, booking workflow and dashboard outputs.

Phase 6: Deployment

The system will be deployed on a secure server so that authorised university users can access it.

Phase 7: Training and Handover

Department secretaries, administrators and planning officers will be trained on how to use the system.

16. Risk Considerations and Mitigation

Risk	Possible Effect	Mitigation
Incomplete room data	System may not reflect all available rooms	Conduct departmental verification before full launch
Users failing to update room status	Dashboard becomes inaccurate	Assign responsibility and include reminders
Double-booking due to manual overrides	Conflicts in room use	Use automated conflict detection
Unauthorised changes	Data integrity problems	Use role-based access and audit trails
Resistance to new system	Low adoption	Provide training and simple user interface
Server downtime	Users cannot access system	Use reliable hosting and backups

17. Expected Outputs

The project will produce:

1. A centralised room inventory database.
2. A web-based room management system.
3. User login and access control.
4. Room booking and approval workflow.
5. Room availability calendar.
6. Usage monitoring dashboard.
7. Departmental and university-wide reports.
8. Audit trail for system activities.
9. User documentation and training guide.

18. Conclusion

The proposed Centralised Room Inventory and Usage Monitoring System will provide the University of Zimbabwe with a reliable digital platform for managing rooms, capacities, schedules and bookings. By bringing all room information into one system, the university will be able to improve room allocation, reduce double-booking, monitor utilisation, and support better planning for lectures, examinations and institutional events.

The system will be simple enough for department secretaries and administrators to update, while also providing useful dashboards for management and planning offices. With proper implementation, training and institutional support, the system can become an important tool for improving space management and operational efficiency at the University of Zimbabwe.